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DATA MINING

MODULE 6 - Clustering

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INTRODUCTION

PRACTICAL OBJECTIVES

1. Students are able to understand the basics of Python programming language
2. Students are able to understand Clustering Algorithm in Python programming language
3. Students are able to understand K-Means and DBSCAN

SOFTWARE / APPLICATION PREPARATION

1. Muslim students may recite the following prayer before starting the lesson

رَضِيتُ بِاللّٰهِ رَبًّا وَبِالْإِسْلَامِ دِينًا وَبِمُحَمَّدٍ نَبِيًّا وَرَسُولًا. رَبِّ زِدْنِي عِلْمًا وَارْزُقْنِي فَهْمًا

2. Hardware

- Laptop/Computer

3. Software (Choose one)

- Visual Studio Code, Google Collaboratory, PyCharm, Jupyter Notebook

4. Tutorial

- Using Google Colab : https://youtu.be/tpDjhSzJor4?si=bFjSP_mf3oB5x520
- Using Google Colab Jupyter in VS Code : <https://youtu.be/HQg13z1w11A?si=RhEY2QY5bir-dzEH>
- Using Jupyter Notebook : https://youtu.be/HQaCF2F_MR0?si=k6ubuGuCg-VS0tQe
- Install Anaconda : https://youtu.be/XyudQ3B30WI?si=7qugN_vTxAM30qDt

5. Google Colab Material

🔗 **MODULE 6 DATA MINING - Clustering.ipynb**

KEYWORDS

Clustering, K-Means, DBSCAN



INTRODUCTION

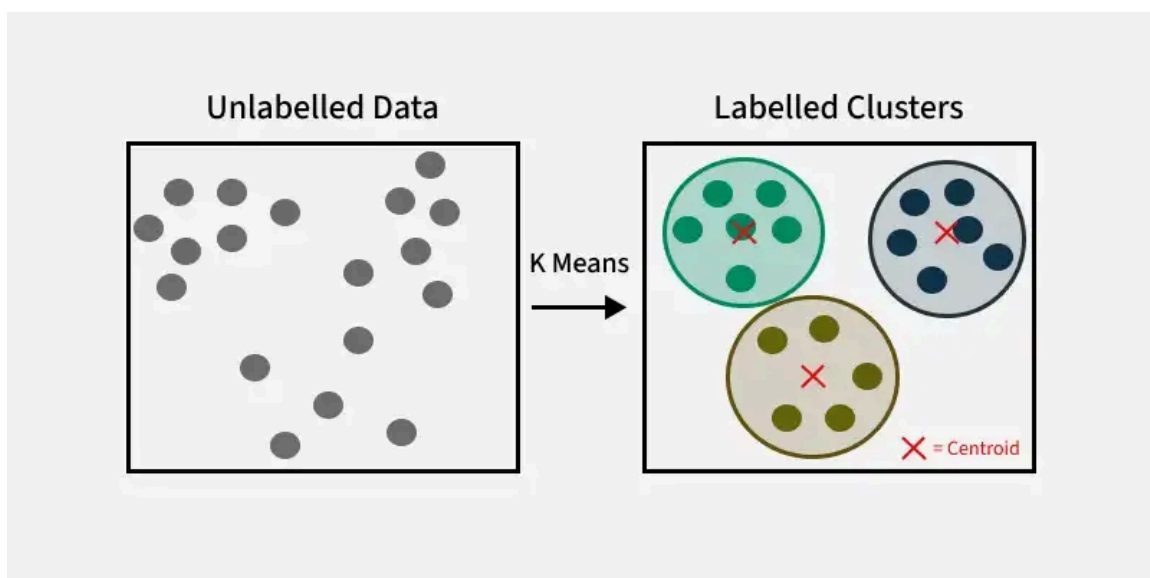
Clustering is a technique in machine learning used to group data into several groups based on similarity or proximity of characteristics. This method is included in unsupervised learning because it does not require labels to separate data.

Clustering has various applications such as customer segmentation, pattern analysis, image recognition, and document grouping. In clustering, the main objective is to minimize the distance between data within a group (intra-cluster) and maximize the distance between groups (inter-cluster).

A. K-Means Clustering

K-Means clustering is a technique used to organize data into groups based on similarities. For example, online shops use K-Means to group customers based on purchase frequency and spending, creating segments such as “Economical Buyers,” “Frequent Buyers,” and “Big Buyers” for personalized marketing.

This algorithm works by first randomly selecting several center points called centroids, and then assigning each data point to the nearest centroid, which forms a cluster. After all points are assigned to clusters, the centroids are updated by finding the average position of the points in each cluster. This process is repeated until the centroids stop changing and form clusters. The purpose of clustering is to divide data points into clusters so that similar data points are included in the same group.



1. K-Means Algorithm Steps

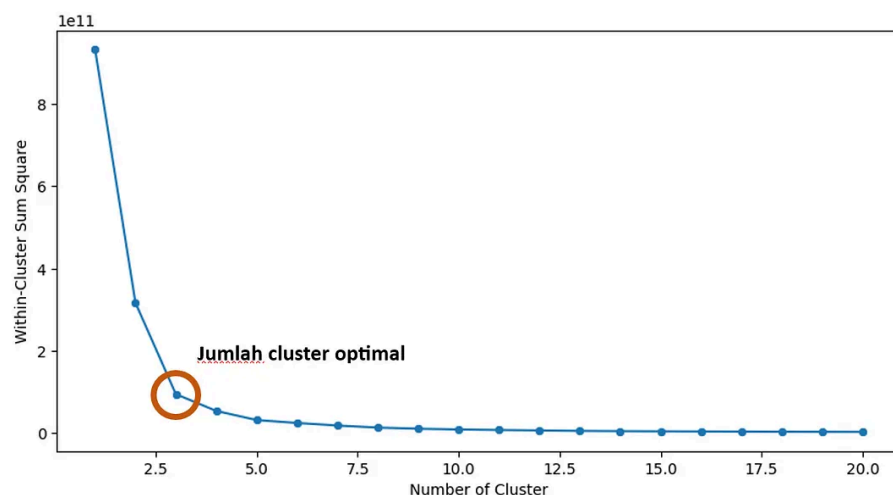
- Determine the number of clusters (K): We determine how many groups we want to create. For example, if we want to group customers into 3 groups, then $K = 3$.
- Initialize random centroids: A centroid is the center point of each group. We select several random points as the initial centroids.
- Group the data: Each data point (e.g., customer) will be grouped into the cluster whose centroid is closest to it. The appeal is that data points that are more similar will be grouped together.
- Update the centroids: After all the data has been grouped, we update the position of the centroids by calculating the average position of the data within each group.
- Repeat steps 3 and 4: This process is repeated several times until the position of the centroids no longer changes or we reach a predetermined iteration limit.

2. Determining the Correct K Value

Although K-Means works well on training datasets, it is important to remember that the weakness of K-Means is that we must determine the number of clusters K before we know that the value of K is optimal. The number of clusters to be selected is not always clear in real-world applications, especially when working with high-dimensional datasets.

a. Elbow Method

This method is an approach used in clustering analysis to determine the optimal number of clusters in the data. This method looks for the point where the decrease in inertia begins to slow down, forming a graph that resembles an elbow shape. If there is an angle or decrease in the graph, then the cluster value can be considered the best.



b. Silhouette Score Method



The silhouette score is a metric used to evaluate how well clustering results in data clustering. This score ranges from -1 to 1, where higher values indicate better clustering results. The value is calculated by measuring the similarity of each data point to its cluster and how different it is from other clusters.

$$S(i) = \frac{b(i) - a(i)}{\max\{a(i), b(i)\}}$$

a = average distance to other data in the same cluster.

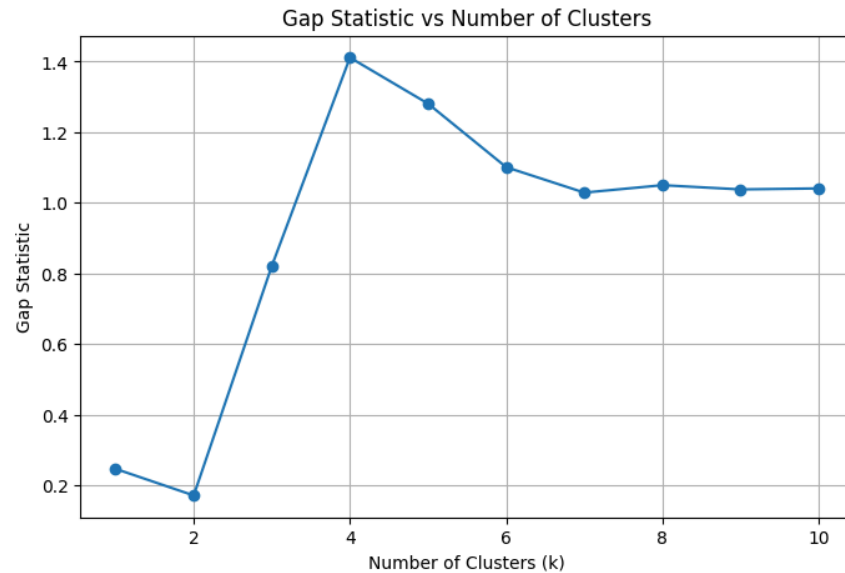
b = average distance to data in the next closest cluster.

Skor Silhouette	Category
$0,7 < skor \leq 1$	Strong Structure
$0,5 < skor \leq 0,7$	Medium Structure
$0,25 < skor \leq 0,5$	Weak Structure
$skor \leq 0,25$	No Structure

c. Gap Statistics Method

Gap Statistics is a method for estimating the optimum number of clusters in cluster analysis. This gap is generated by comparing the variation between the original data and the random data. A high gap value indicates the optimal number of clusters. This helps us understand whether the patterns we see in the data are truly meaningful or just coincidental.





**In the figure, the optimal K value based on the gap obtained is 3.*

3. Case Study: Pharmaceutical Product Segmentation Based on Sales

In this section, we will use K-Means to segment pharmaceutical products based on their sales patterns. The dataset used is OTC (Over-The-Counter) sales data from pharmacies.

Business Scenario:

A pharmaceutical company wants to understand the sales patterns of its OTC products to:

- Identify products with similar sales performance
- Optimize marketing strategies based on product groups
- Determine more efficient inventory stock

A. Dataset Preparation

Let's download and load the dataset first.

```
# Download dataset dari Kaggle
path =
kagglehub.dataset_download("atharvasoundankar/pharmacy-otc-sales-data-2022")
print("Path to dataset files:", path)

# List semua file dalam direktori untuk melihat struktur folder
```

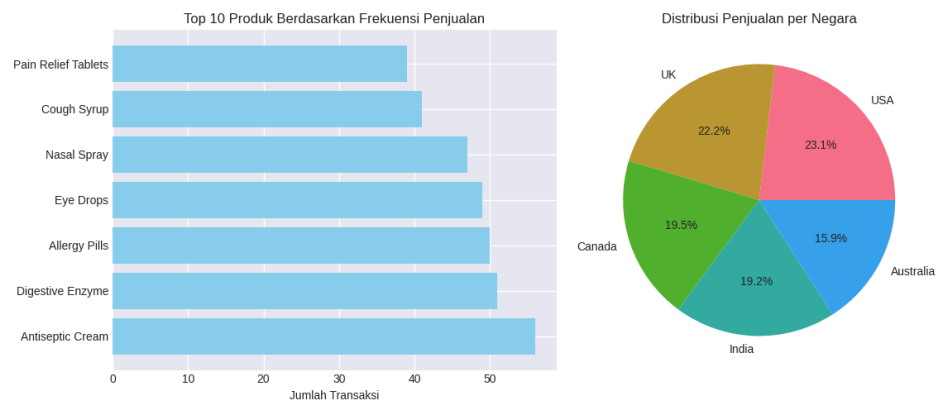


```
print("\n=== Struktur Folder Dataset ===")
for root, dirs, files in os.walk(path):
    level = root.replace(path, '').count(os.sep)
    indent = ' ' * 2 * level
    print(f'{indent}{os.path.basename(root)}/')
    subindent = ' ' * 2 * (level + 1)
    for file in files:
        print(f'{subindent}{file}')
```

B. Data Exploration and Analysis (EDA)

Before performing clustering, we first need to understand the characteristics of the data.

Output :



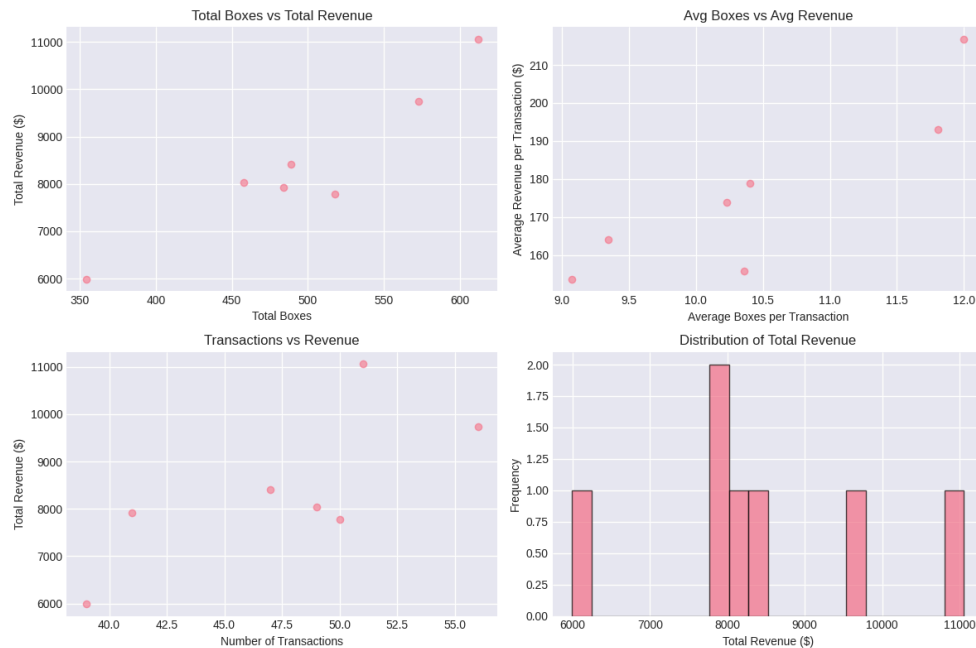
C. Data Preparation for Clustering

For K-Means, we will use the aggregate features of each product:

- Total boxes shipped
- Average boxes per transaction
- Total revenue
- Average revenue per transaction
- Number of transactions

Output :





D. Data Normalization

K-Means is sensitive to data scale, so we need to normalize it using StandardScaler.

Output :

```
=== Data Setelah Normalisasi (5 data pertama) ===
  Total_Boxes  Avg_Boxes  Num_Transactions  Total_Revenue  Avg_Revenue
0    0.255628  -0.097993    0.445827    -0.428984   -1.006285
1    0.968794  -0.222370    1.547283     0.889021   -0.130107
2   -0.185238   1.307565   -1.206356   -0.338295    0.795392
3    1.474494   1.497377    0.629403    1.779742    1.935614
4   -0.522371  -1.083485    0.262251   -0.257779   -0.604259

=== Statistik Data Ternormalisasi ===
   Total_Boxes  Avg_Boxes  Num_Transactions  Total_Revenue  Avg_Revenue
count         7.00      7.00           7.00           7.00           7.00
mean           0.00      0.00           0.00          -0.00           0.00
std           1.08      1.08           1.08           1.08           1.08
min          -1.87     -1.35          -1.57          -1.64          -1.10
25%          -0.35     -0.65          -0.66          -0.38          -0.81
50%          -0.12     -0.10           0.26          -0.26          -0.13
75%           0.61      0.63           0.54           0.44           0.45
max           1.47      1.50           1.55           1.78           1.94
```

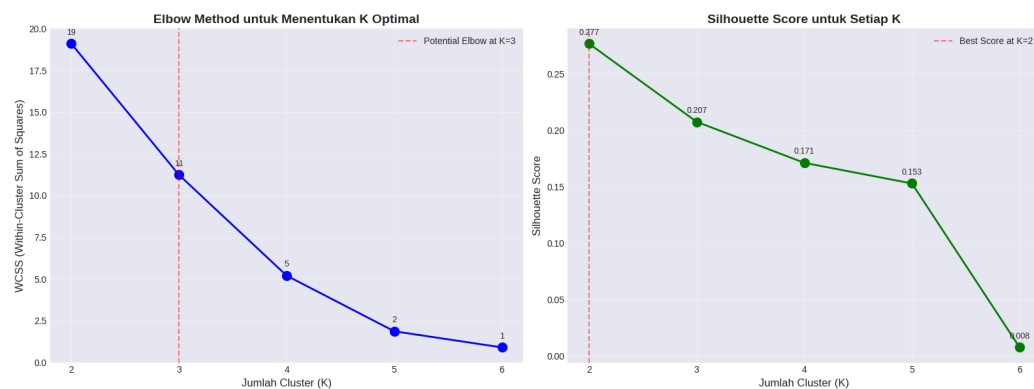
E. Finding the Best "Number of Groups" (Elbow Method & Silhouette Score)

Imagine you're organizing a pharmacy. You want to group your existing products, but you're unsure how many categories to divide them into.



- Elbow Method: Think of it like pulling a rubber band. The more groups there are, the closer the products within the group become (the smaller the WCSS). We're looking for the "elbow" point where adding a new group no longer provides significant benefits.
- Silhouette Score: This is the product's "comfort" score within its group. The higher the score (closer to 1), the more the product feels "fit" within its group and is significantly different from other groups. A low score means the product feels "uneasy" because it's similar to its neighboring groups.

Output :



Potential Elbow Point: K = 3

Best Silhouette Score: K = 2 (Score = 0.2767)

Analysis Results Explanation:

Based on the graph and calculations, we obtain the following picture:

1. Data Validation: We have 7 unique products to group. In K-Means, the number of groups (K) cannot be equal to or greater than the number of data points, so we tested a range of K from 2 to 6.
2. Elbow Method Results: The graph shows a drastic drop at K=3. This point is considered the "elbow," indicating that dividing the data into 3 groups is quite efficient in summarizing product similarities.
3. Silhouette Score Results: The highest score was found at K=2 (0.2767). Although statistically K=2 is the most stable, in practice, we often combine this result with the Elbow Method.
4. Conclusion: Based on the analysis summary, we have two strong candidates. We can choose K=2 if we want highly contrasting groupings, or K=3 if we want a slightly more detailed but still efficient division.



F. Implementing K-Means with Optimal K

After we previously "queried" the data using the Elbow Method, now it's time to actually divide the pharmaceutical products into their respective groups.

Imagine we're attaching color labels to each product: Products with similar sales performance will receive the same color label. Here, we use

$$K=3$$

as the optimal number of groups to see the differences between high-selling, average-selling, and low-selling products.

Output :

```

=== Hasil Clustering dengan K=3 ===
  Product  Total_Boxes  Avg_Boxes  Num_Transactions  \
0  Allergy Pills         518   10.360000          50
1  Antiseptic Cream       573   10.232143          56
2  Cough Syrup           484   11.804878          41
3  Digestive Enzyme       612   12.000000          51
4  Eye Drops             458    9.346939          49
5  Nasal Spray           489   10.404255          47
6  Pain Relief Tablets    354    9.076923          39

  Total_Revenue  Avg_Revenue  Cluster
0    7782.58    155.651600         1
1    9736.28    173.862143         1
2    7917.01    193.097805         1
3   11056.61    216.796275         2
4    8036.36    164.007347         1
5    8407.36    178.880000         1
6    5993.09    153.668974         0

=== Jumlah Produk per Cluster ===
Cluster
0    1
1    5
2    1
Name: count, dtype: int64

=== Karakteristik Setiap Cluster ===
  Total_Boxes  Avg_Boxes  Num_Transactions  Total_Revenue  Avg_Revenue
Cluster
0           354.0      9.08              39.0      5993.09      153.67
1           504.4     10.43              48.6     8375.92      173.10
2           612.0     12.00              51.0     11056.61      216.80
  
```

Explanation of Output Results:

Based on the execution results above, the algorithm has divided products into three clusters (0, 1, and 2) with the following characteristics:

- Cluster 0 (Economy Products): Contains only one product (Pain Relief Tablets). This cluster has the lowest sales figures across all categories,



both in terms of the number of boxes sold and revenue. This is the product group with the lowest performance.

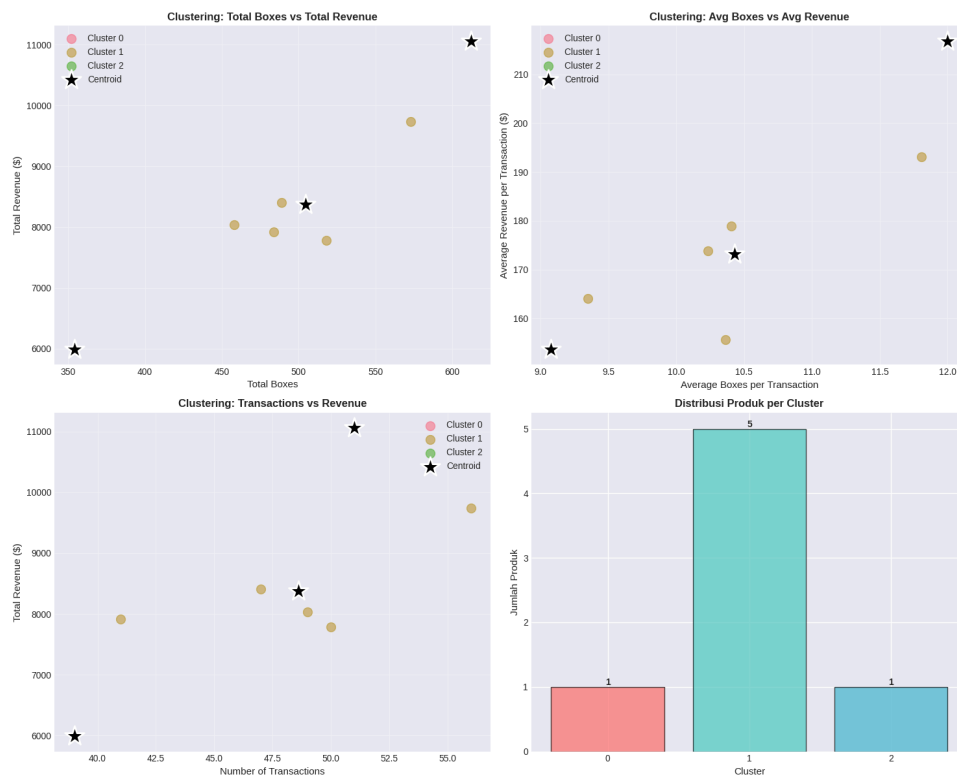
- Cluster 1 (Mainstream Products): Contains the majority of products (5 products). This is the store's "backbone" group with stable sales and revenue performance at a medium level.
- Cluster 2 (Sultan Products): Contains only one product (Digestive Enzyme). This product is the star product because it has the highest total sales (612 per box) and contributes the largest revenue (over 11,000).

In short: This clustering successfully separates products based on their sales "strength," allowing pharmacy owners to identify which products need to be promoted more intensively (Cluster 0) and which ones should be kept in stock because they are the most profitable (Cluster 2).

G. Visualizing Clustering Results

After calculating the numbers in the previous step, we will now translate the data into visual coordinates. This visualization is like a "location map" for our pharmaceutical products. By looking at the position of the dots and their color distribution, we can immediately see which groups are market leaders and which are lagging behind without having to bother looking at tables of numbers.

Output :



Based on the four graphs produced, here are the key points:

- Scatter Plots: The first three graphs show the relationship between the number of items sold, the number of transactions, and revenue. You can see that there are three distinct colors that are quite clearly separated. The Black Star Point (Centroid) is the "class leader" or center of each cluster.
- Cluster 2 (Green Dot in the Top Right): This is the top-performing product. Its position is always in the top right corner, meaning it has the most transactions and generates the most revenue.
- Cluster 0 (Red Dot in the Bottom Left): This is the least popular product. Its position is in the bottom left corner because its sales are the lowest compared to its peers.
- Cluster 1 (Yellow Dot in the Middle): This is the majority group. They are clustered in the middle area, indicating these products are "safe" items with stable and consistent sales.

Product Distribution (Bar Chart): The bar chart in the bottom right shows the distribution of our population. We can clearly see that the majority of products (5 products) fall into the middle category (Cluster 1), while the bottom and top categories each have only 1 exclusive product.

H. Cluster Interpretation and Profiling

Data without narrative is just a bunch of numbers. In this step, we perform "profiling," which involves assigning an identity or label to each cluster based on its purchasing behavior. We will summarize the average value for each cluster and assign appropriate labels so that the results of this analysis can be used by pharmacy management to make real-world business decisions.

Output :



```
=====
PROFILING CLUSTER
=====

CLUSTER 0
=====
Jumlah Produk: 1

Karakteristik:
- Total Boxes (avg)      : 354.00
- Avg Boxes/Transaction  : 9.08
- Total Transactions      : 39.00
- Total Revenue (avg)    : $5993.09
- Avg Revenue/Transaction: $153.67

Top 5 Produk dalam Cluster 0 berdasarkan Revenue:
- Pain Relief Tablets      : $ 5993.09 (354 boxes)

=====
CLUSTER 1
=====
Jumlah Produk: 5

Karakteristik:
- Total Boxes (avg)      : 504.40
- Avg Boxes/Transaction  : 10.43
- Total Transactions      : 48.60
- Total Revenue (avg)    : $8375.92
- Avg Revenue/Transaction: $173.10

Top 5 Produk dalam Cluster 1 berdasarkan Revenue:
- Antiseptic Cream        : $ 9736.28 (573 boxes)
- Nasal Spray             : $ 8407.36 (489 boxes)
- Eye Drops               : $ 8036.36 (458 boxes)
- Cough Syrup             : $ 7917.01 (484 boxes)
- Allergy Pills           : $ 7782.58 (518 boxes)

=====
CLUSTER 2
=====
Jumlah Produk: 1

Karakteristik:
- Total Boxes (avg)      : 612.00
- Avg Boxes/Transaction  : 12.00
- Total Transactions      : 51.00
- Total Revenue (avg)    : $11056.61
- Avg Revenue/Transaction: $216.80

Top 5 Produk dalam Cluster 2 berdasarkan Revenue:
- Digestive Enzyme        : $ 11056.61 (612 boxes)

=====
Hasil clustering telah disimpan di /content/product_clusters.csv
=====
```

Final Conclusion of the K-Means Test: From the profiling results above, we can simplify our findings into three main profiles:

- Cluster 0 (Low Performer): This group contains the lowest-performing products. For example, Pain Relief Tablets. This product is rarely traded and generates the lowest revenue. Business Advice: A promotional strategy or inventory evaluation is needed to prevent inventory from piling up in the warehouse.
- Cluster 1 (Medium Performer): This is the "balance" group containing the largest number of products (such as Antiseptic Cream and Cough Syrup). Their performance is stable and average. Business Advice: Maintain stock availability as this group maintains the pharmacy's cash flow.
- Cluster 2 (High Performer): The "Star" group currently consists of Digestive Enzyme. This product is a leader in both sales volume and



revenue. Business Advice: Ensure stock availability (never run out) and provide special treatment to maintain customer loyalty.

With this step completed, we have successfully transformed messy transaction data into coherent strategic information. The final results have also been saved in the product_clusters.csv file which is ready to be presented or processed further.

B. DBSCAN Clustering

Density-Based Spatial Clustering of Applications with Noise (DBSCAN) is a density-based clustering algorithm that groups nearby data points and marks outliers as noise based on their density in feature space. **DBSCAN clusters data based on how densely the points are located to each other.**

Unlike K-Means or other hierarchical clustering algorithms that assume clusters are compact and spherical, DBSCAN excels at handling the irregularities of real-world data such as:

- Arbitrary-shaped clusters: Clusters can be any shape, not just circular or convex.
- Noise and Outliers: Effectively identifies and handles noise points without throwing them into other clusters.



DBSCAN Algorithm Steps:

1. Determine the parameters ϵ (radius) and min_samples (minimum number of points within the radius).
2. Start with the initial data points, check whether the number of points within the ϵ radius meets min_samples .
 - If yes, make this point the core point and form a cluster.
 - If not, mark it as noise or an outlier.



- Repeat the process for all points until there are no more points that can be included in the cluster.

Description:

- eps (epsilon): The maximum distance between two points to be considered "close neighbors."
- minPts: The minimum number of neighbors (including itself) for a point to be called a core point.
- Core Point: A point that has enough neighbors ($\geq \text{minPts}$).
- Border Point: A point within the eps range of a core point, but with fewer than minPts neighbors.
- Noise Point (Outlier): A point that is neither a core point nor a border point – considered to belong to no cluster.

Case Study: Anomalous Pattern Detection in Pharmaceutical Sales

DBSCAN is well-suited for detecting outliers and irregularly shaped clusters. In this case, we will use DBSCAN to:

- Identify unusual (anomalous) transaction patterns
- Group transactions based on data density
- Find products with unique sales patterns

A. Data Preparation for DBSCAN

```
=== Preview Data untuk DBSCAN ===
```

	Date	Product	Sales Person	Boxes Shipped	Amount (\$)	\
0	2022-02-06	Pain Relief Tablets	Divya Mehra	7	67.43	
1	2022-02-05	Antiseptic Cream	Nikhil Batra	18	421.46	
2	2022-02-19	Cough Syrup	Ritika Shah	2	48.25	
3	2022-03-28	Pain Relief Tablets	Nikhil Batra	5	59.89	
4	2022-01-17	Digestive Enzyme	Ritika Shah	8	147.73	

	Country	Month	DayOfWeek	Revenue_per_Box
0	Australia	2	6	9.632857
1	Canada	2	5	23.414444
2	UK	2	5	24.125000
3	UK	3	0	11.978000
4	USA	1	0	18.466250

Total Transaksi: 333

B. Determining DBSCAN Parameters (eps and min_samples)

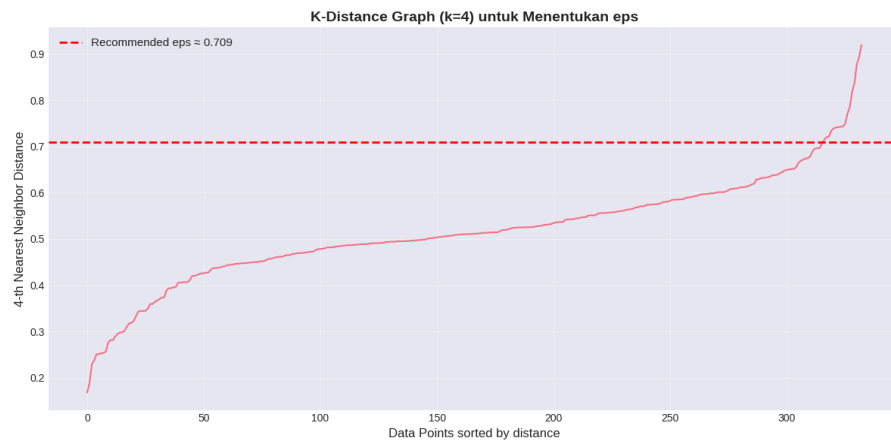
While K-Means requires us to specify the number of clusters upfront, DBSCAN requires us to determine how close together data must be to be considered "clusters."



- Epsilon (eps): Think of it as a radar radius. If the radar is set too small, all data will be considered isolated (noise). If it is too large, all data will be considered one large cluster.
- Min_samples: The minimum number of members required for a cluster to be officially called a "Cluster."

To find the right radius (eps), we use a K-Distance Graph. Imagine each data point finding its four nearest neighbors, then we order their distances from closest to furthest.

Output :



Based on the resulting graph (second image), we can see the search process for the ideal eps value:

- Finding the "Jump Point": We observe the pink line slowly rising. The most ideal point is when the graph begins to climb sharply (the "elbow" area).
- Recommended Value: Based on the system's calculations at the 95th percentile, the eps value is found to be approximately 0.709. At this point, the distance between the data points begins to widen, making it a suitable radius boundary to separate "cluster centers" from "isolated data."
- Data Density: By setting min_samples = 4, we instruct the algorithm that a new point can be considered a cluster center if it has at least 4 neighbors within the 0.709 radius.

C. Experiment with Different DBSCAN Parameters

Finding the right DBSCAN parameters is like adjusting the focus on a camera lens. If the settings aren't right, the image can be too pixelated (too many clusters) or too blurry (everything is considered one cluster).



Since we can't guess the exact numbers, we conducted Parameter Experimentation. We tried various combinations of eps (radius) and min_samples (minimum member requirement) to see which produced the most logical and clean clustering.

Output :

```
=====
EKSPERIMEN PARAMETER DBSCAN
=====
eps=0.3, min_samples=3: Clusters=21, Noise=248 (74.5%), Silhouette=0.593
eps=0.3, min_samples=4: Clusters=9, Noise=292 (87.7%), Silhouette=0.658
eps=0.3, min_samples=5: Clusters=2, Noise=322 (96.7%), Silhouette=0.915
eps=0.3, min_samples=6: Clusters=0, Noise=333 (100.0%), Silhouette=-1.000
eps=0.5, min_samples=3: Clusters=15, Noise=68 (20.4%), Silhouette=-0.103
eps=0.5, min_samples=4: Clusters=15, Noise=112 (33.6%), Silhouette=0.199
eps=0.5, min_samples=5: Clusters=13, Noise=160 (48.0%), Silhouette=0.236
eps=0.5, min_samples=6: Clusters=12, Noise=213 (64.0%), Silhouette=0.402
eps=0.7, min_samples=3: Clusters=1, Noise=1 (0.3%), Silhouette=-1.000
eps=0.7, min_samples=4: Clusters=1, Noise=3 (0.9%), Silhouette=-1.000
eps=0.7, min_samples=5: Clusters=1, Noise=4 (1.2%), Silhouette=-1.000
eps=0.7, min_samples=6: Clusters=2, Noise=6 (1.8%), Silhouette=0.048
eps=0.9, min_samples=3: Clusters=1, Noise=0 (0.0%), Silhouette=-1.000
eps=0.9, min_samples=4: Clusters=1, Noise=0 (0.0%), Silhouette=-1.000
eps=0.9, min_samples=5: Clusters=1, Noise=0 (0.0%), Silhouette=-1.000
eps=0.9, min_samples=6: Clusters=1, Noise=0 (0.0%), Silhouette=-1.000
eps=1.1, min_samples=3: Clusters=1, Noise=0 (0.0%), Silhouette=-1.000
eps=1.1, min_samples=4: Clusters=1, Noise=0 (0.0%), Silhouette=-1.000
eps=1.1, min_samples=5: Clusters=1, Noise=0 (0.0%), Silhouette=-1.000
eps=1.1, min_samples=6: Clusters=1, Noise=0 (0.0%), Silhouette=-1.000

=== Top 5 Kombinasi Parameter (berdasarkan Silhouette Score) ===
eps  min_samples  n_clusters  noise_ratio  silhouette
2  0.3           5           2    96.696697    0.915500
1  0.3           4           9    87.687688    0.657983
0  0.3           3          21    74.474474    0.593405
7  0.5           6          12    63.963964    0.401803
6  0.5           5          13    48.048048    0.236074
```

From the "Top 5 Combinations" table above, there are several interesting things we can learn:

- Cluster vs. Noise Balance: When we use eps=0.3 with min_samples=5, we get a very high Silhouette Score (0.915). However, notice the Noise Ratio: almost 96%! This means that the system considers almost all data to be "garbage" or anomalous due to our strict rules.
- Reading Noise: A Noise value (marked with a -1 in the data) indicates data that doesn't fit into any clusters. If there's too much noise (as in the first row of the top 5 results), it means our radius is too small or the membership requirements are too stringent.
- Selecting the Best: In the real world, we not only look for the highest Silhouette Score, but also for a reasonable number of clusters and a level of noise that doesn't overpower the clusters.

Parameter Analysis:

- As eps increases, the number of clusters tends to decrease because the radar radius widens, resulting in more points being merged.
- As min_samples increases, the amount of noise increases because the requirements for becoming an "official group" become more difficult to



fulfill.

D. Implementing DBSCAN with Optimal Parameters

After conducting various simulations, we now reach the decision stage. We select the parameters with the highest scores to "lock" our clustering model. Like choosing a pair of glasses, we've found the clearest lenses to distinguish between truly solid data clusters and those that are merely "noise" or anomalies.

Output :

```

=== Parameter Optimal yang Dipilih ===
eps = 0.3
min_samples = 5

=== Hasil DBSCAN Clustering ===
Jumlah Cluster: 2
Jumlah Noise Points: 322 (96.70%)

Distribusi Data per Cluster:
-1    322
 0      5
 1      6
Name: count, dtype: int64

```

Based on the final execution results, there are several interesting points we can learn from how DBSCAN works:

- **Parameters Used:** The system selected $\text{eps}=0.3$ and $\text{min_samples}=5$. This is a very strict setting, where a point must have at least 5 neighbors within a very narrow radius to be considered a cluster.
- **Noise Dominance (Label -1):** There are 322 points considered noise. In general terms, DBSCAN assesses that most of our transaction data has patterns that are too unique or scattered to be placed into any large clusters.
- **Formation of "Core" Clusters:** Despite much discarded data, DBSCAN successfully identified two main clusters:
- **Cluster 0:** Contains 5 products with very similar patterns.
- **Cluster 1:** Contains 6 products that are also tightly clustered.

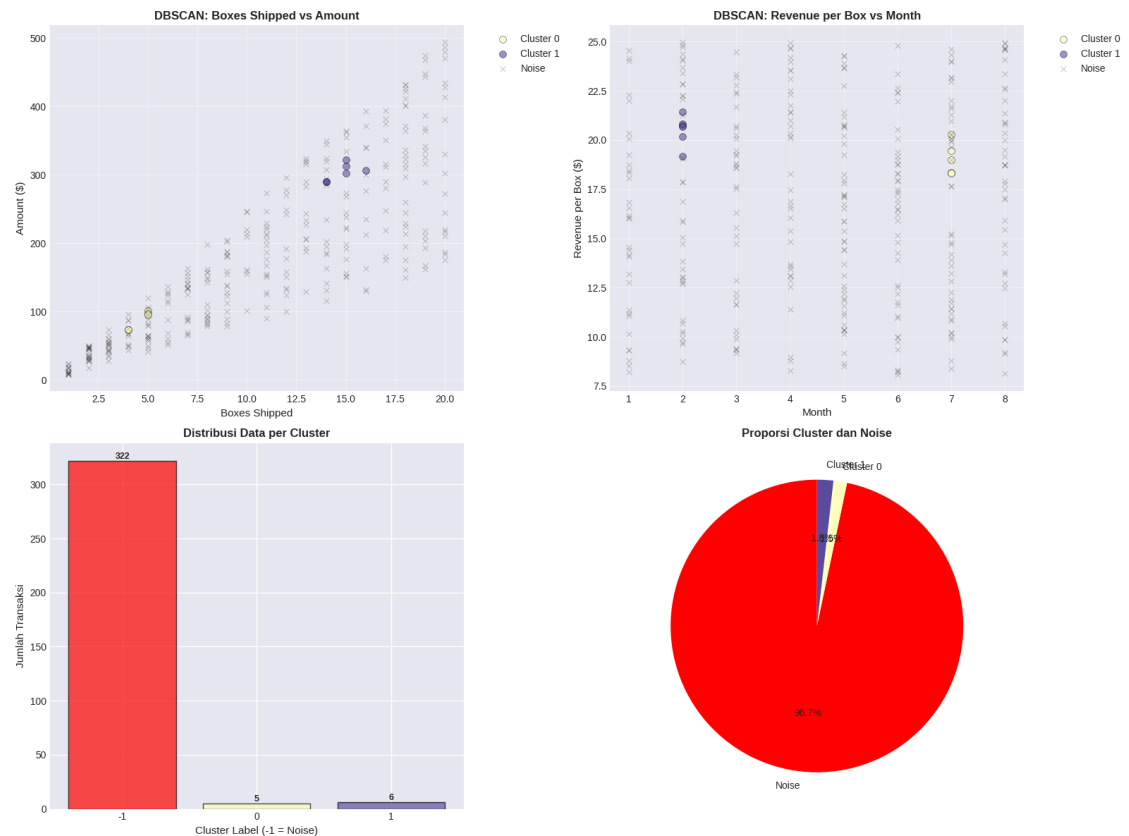
Conclusion: Unlike K-Means, which forces all data into clusters, DBSCAN is much more "honest." It only takes truly similar data (Clusters 0 and 1) and boldly marks different data as anomalies (Noise).



E. Visualizing DBSCAN Results

While K-Means forces all data into clusters, this DBSCAN visualization presents a different picture. This visualization helps us clearly distinguish which data are "dense clusters" (Clusters) and which are "outliers" or anomalies (Noise). This is a great way to see how the epsilon and min_samples parameters we chose earlier perform in practice.

Output :



Based on the graph above, we can take several key points:

- **Noise Dominance (Red/Cross Marks):** The pie chart and bar chart show a striking fact: 96.7% of the data (322 transactions) is considered noise. In the context of this pharmacy data, this means that most transactions have very diverse and unique patterns, preventing them from forming a tight cluster pattern.
- **Formed Clusters (Yellow & Purple Dots):** DBSCAN only managed to identify two small, solid clusters.
 - **Cluster 0 (Yellow):** A small cluster that appears to be tightly clustered in an area of low transaction volume.



- Cluster 1 (Purple): A small cluster characterized by medium-level transactions.
- Data Distribution: In the scatterplot (above), the widely scattered gray cross dots represent transactions that are considered to be out of step with the general trend. This is very useful for pharmacies in identifying unusual or "special case" transactions.

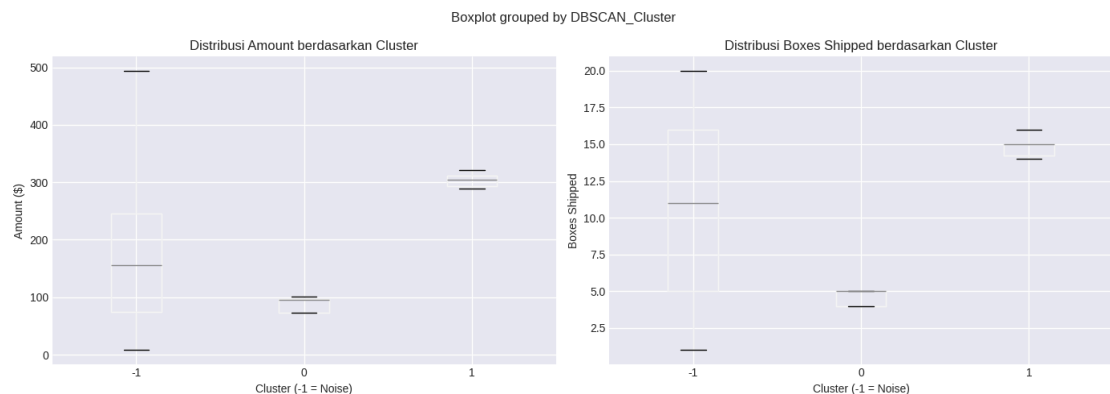
In short, DBSCAN tells us that this transaction data is highly variable. Instead of dividing everything into large groups, DBSCAN prefers to show us which points actually have extreme similarities, and discards the rest as anomalies.

F. In-depth Analysis of Noise and Outliers

In the DBSCAN algorithm, data that doesn't fit into any of the categories (Noise) isn't garbage, but rather valuable information. This data is unique transactions that deviate from the usual trend.

Imagine sifting through gold; Noise are large nuggets that don't pass through the sieve due to their unusual size. In this step, we'll examine the characteristics of these 322 "unique" transactions to see if they are suspicious transactions, large-scale special orders, or simply data anomalies.

Output :



Explanation of Analysis Results: Based on the boxplot and statistical table generated, we can understand the behavior of the Noise data as follows:

- Very Wide Range: If you look at the boxplot graph (long white box), the Noise data (label -1) has a very wide range of Amount and Boxes Shipped values, ranging from the smallest to the most astronomical (\$494). This is



in stark contrast to Clusters 0 and 1, whose boxes are very small and stable.

- Who are the "Major Players" in the Noise?: The Top 10 Transactions table shows that the majority of the noise is high-volume transactions (18-20 boxes). For example, the sale of Cough Syrup for \$494 by Rajesh Patel. Because this value is far above the average, DBSCAN considers this not a typical daily transaction, but rather an outlier.
- Anomaly is an Opportunity: From a business perspective, this Noise data represents a "Whale" transaction (a large purchase). Although statistically considered an anomaly, for a pharmacy, this is the most profitable transaction.
- Visual Conclusion: The boxplot graph proves that DBSCAN is very smart in separating "uniform normal data" (Cluster 0 & 1) from "varied unique data" (Noise).

G. Profiling Each DBSCAN Cluster

After separating the "normal" data from the "outliers," we will now assign labels or identifiers to each cluster. This profiling is crucial because it helps us understand the specific behavior of each cluster, from which products are purchased most frequently to which countries are most dominant. With this information, marketing or operational strategies can be tailored to each cluster.

Output :



```

=====
CLUSTER 0
=====
Jumlah Transaksi: 5

Karakteristik Rata-rata:
- Boxes Shipped : 4.60
- Amount : $88.07
- Revenue/Box : $19.08

Produk Dominan dalam Cluster:
- Pain Relief Tablets : 2 transaksi (40.0%)
- Antiseptic Cream : 2 transaksi (40.0%)
- Eye Drops : 1 transaksi (20.0%)

Negara Dominan:
- India : 1 transaksi (20.0%)
- UK : 1 transaksi (20.0%)
- Canada : 1 transaksi (20.0%)

=====
CLUSTER 1
=====
Jumlah Transaksi: 6

Karakteristik Rata-rata:
- Boxes Shipped : 14.83
- Amount : $303.85
- Revenue/Box : $20.50

Produk Dominan dalam Cluster:
- Eye Drops : 2 transaksi (33.3%)
- Digestive Enzyme : 2 transaksi (33.3%)
- Pain Relief Tablets : 1 transaksi (16.7%)

Negara Dominan:
- UK : 3 transaksi (50.0%)
- Canada : 2 transaksi (33.3%)
- USA : 1 transaksi (16.7%)

```

Profiling Results Explanation:

Based on the report above, we can divide the transaction data into three major profiles:

- **NOISE POINTS (Specialists):** Contains the largest transaction group (322 transactions). Interestingly, despite being called "noise," this group includes popular products like Antiseptic Cream and Allergy Pills, with primary markets in the US and UK. This group is called noise not because it is unimportant, but because the transaction values vary so much that they do not form a narrow cluster pattern.
- **CLUSTER 0 (Small & Measured Transactions):** This group contains transactions in small quantities (average 4.6 boxes) with a value of around \$88. The dominant products are Pain Relief Tablets and Antiseptic Cream. This is a profile of small retail transactions evenly distributed across various countries.
- **CLUSTER 1 (Premium Transactions):** This is the "High-Value" group. The number of boxes is larger (average 14.8 boxes) and the monetary value is



much higher (average \$303). The standout products were Eye Drops and Digestive Enzyme, with the UK being the most dominant market (50%).

The conclusion of this experiment was that DBSCAN successfully isolated two highly consistent transaction patterns (Clusters 0 and 1) amidst a sea of highly diverse transaction data (Noise).

C. Case Study Conclusion

From the analysis using K-Means and DBSCAN on the pharmaceutical sales dataset:

K-Means yields:

- 3 clear product segments: Low, Medium, and High Performers
- All products are categorized for strategic planning
- Easy to use for inventory management and sales forecasting

DBSCAN yields:

- Identifies anomalous transactions (noise points) that require investigation
- Clusters based on density patterns in transactions
- Useful for detecting unusual sales patterns

Implementation Recommendations:

1. Use K-Means for product segmentation and strategic planning
2. Use DBSCAN for transaction monitoring and anomaly detection
3. Combine both methods for comprehensive analysis
4. Update the model regularly with the latest data

D. Comparison of K-Means vs DBSCAN

As a comprehensive conclusion to this module, we will compare the two algorithms we've learned. Understanding the differences between K-Means and DBSCAN is crucial for choosing the right tool based on your data characteristics and desired analysis goals.

Here's an in-depth comparison table to help you understand the advantages and disadvantages of each method in a real-world data processing context:



Aspects	K-Means	DBSCAN
Basic Principles	Clustering data based on distance to centroid	Clustering data based on density
Number of Clusters	Determined in advance (e.g., $k = 3$)	Determined automatically from data density
Data Assignment	All data points must be included in a cluster	Not all data points are included in a cluster (can be noise)
Outlier Handling Ability	Weak, very sensitive to outliers	Strong, outliers are marked as noise
Cluster Shape	Spherical	Arbitrary
Main Parameters	Number of clusters (k)	eps and min_samples
Advantages	- Simple and fast - Easy to interpret - Suitable for product segmentation	- Can detect anomalies - No need to determine the number of clusters - Robust to noise
Disadvantages	- Must determine k - Sensitive to outliers - Less effective for complex shapes	- Sensitive to parameter selection - Difficult at varying densities - Less optimal in high dimensions
Suitable for Use	Customer or product segmentation	Fraud detection, anomaly detection, quality control
Case Examples	Product grouping based on price and sales	Detecting unusual transactions or extreme data
Analysis Objectives	Categorization and business strategy	Identifying abnormal patterns

CODELAB (20%)

[DATASET CODELAB](#)

The code below is a framework for clustering. Please correct and complete it to ensure it runs error-free.

1. Import the required library (10 Poin)

```
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.preprocessing import ...
from sklearn.cluster import ...
from sklearn.compose import ...
```

2. Display the dataset (10 Point)

```
df = pd.read_csv('your_file_path.csv')
```

3. Perform EDA as necessary (15 Point)

```
# Write Code in this cell
```

4. Create data visualizations (15 Point)

```
# Write Code in this cell
```

5. Preprocessing Data (15 Point)

```
# Tips: Gunakan pd.get_dummies() untuk one hot encoding
df_encoded = pd.get_dummies(df, columns=['...'])
```

```
#Normalisasi Data: Usia, Pendapatan Tahunan, dan Skor Pengeluaran
scaler = MinMaxScaler()
df_encoded[['...', '...', '...']] =
scaler.fit_transform(df_encoded[['...', '...', '...']])
```

6. K-Means Application (25 Point)

```
# Gunakan metode Elbow untuk menentukan jumlah kluster yang ideal
wcss = []
for i in range(1, 11):
    kmeans = KMeans(n_clusters=i, init='k-means++', random_state=42)
    kmeans.fit(df_encoded[['...', '...']])
    wcss.append(kmeans.inertia_)
```



```
plt.plot(range(1, 11), wcss)
plt.title('Metode Elbow')
plt.xlabel('Jumlah Kluster')
plt.ylabel('WCSS')
plt.show()
```

```
# Lanjutkan proses clustering
```

7. Cluster Interpretation (10 Point)

After determining the number of clusters, perform an analysis to understand the characteristics of each cluster.

LAB ASSIGNMENT (80%)

You will perform clustering analysis on the dataset to group patients based on their characteristics. Use the K-Means and DBSCAN methods to find patterns in the data.

ODD NIM

[LAB ASSIGNMENT DATASET FOR ODD NIM](#)

You will perform clustering analysis on the Covid-19 dataset in Indonesia. Use the K-Means and DBSCAN methods to find patterns in the data.

1. Data Preprocessing: (15%)

- Download and load the dataset into Python using pandas. (20 Point)
- Explore the dataset by displaying the first 5 data points and data type information for each column. (20 Point)
- Check for missing values in the dataset and handle them with an appropriate strategy. (20 Point)
- Encode the required categorical variables. (20 Point)
- Normalize the numerical features that will be used in clustering. (20 Point)

2. K-Means Clustering (20%)

- Use the Elbow, Silhouette Score, and Gap Statistics methods to determine the optimal number of clusters (K value). (25 Point)
- Apply the K-Means Clustering algorithm to the dataset with the determined number of clusters. (25 Point)
- Visualize the clustering results using a scatter plot with two main features. (25 Point)



- Interpret the clustering results. (25 Point)

3. DBSCAN Clustering (20%)

- Apply the DBSCAN algorithm to the dataset and determine the appropriate eps and min_samples parameters using a systematic approach. (40 Point)
- Visualize the clustering results using a scatter plot. (20 Point)
- Compare the clustering results between K-Means and DBSCAN. Explain the advantages and disadvantages of each method in the context of this dataset. (40 Point)

4. Explain to the Assistant (25%)

EVEN NIM

LAB ASSIGNMENT DATASET FOR EVEN NIM

You are asked to perform clustering analysis on the dataset. The purpose of this analysis is to group individuals based on their characteristics and identify patterns that may be associated with the risk of heart attack.

1. Data Preprocessing: (15%)

- Download and load the dataset into Python using pandas. (20 Point)
- Explore the dataset by displaying the first 5 data points and the data type information for each column. (20 Point)
- Check for missing values in the dataset and handle them with an appropriate strategy. (20 Point)
- Perform encoding on the required categorical variables. (20 Point)
- Normalize the numerical features that will be used in clustering. (20 Point)

2. K-Means Clustering (20%)

- Use the Elbow method, Silhouette Score, and Gap Statistics to determine the optimal number of clusters (K value). (25 Point)
- Apply the K-Means Clustering algorithm to the dataset with the determined number of clusters. (25 Point)
- Visualize the clustering results using a scatter plot with two main features. (25 Point)
- Interpret the clustering results. (25 Point)

3. DBSCAN Clustering (20%)

- Apply the DBSCAN algorithm to the dataset and determine the appropriate eps and min_samples parameters using a systematic approach. (40 Point)
- Visualize the clustering results using a scatter plot. (20 Point)



- Compare the clustering results between K-Means and DBSCAN. Explain the advantages and disadvantages of each method in the context of this dataset. (40 Point)

4. Explain to the Assistant (25%)

GRADING CRITERIA & DETAILS

Activity	Description	Weight
Codelab		10%
Lab Assignment	Data Preprocessing	15%
	K-Means Clustering	15%
	DBSCAN Clustering	15%
	Explanation to Assistant	45%
Sub Total		90%
Total		100%

